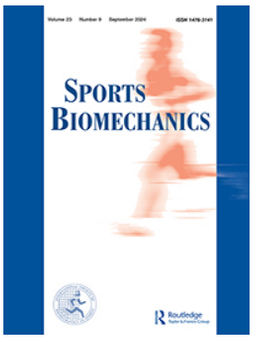

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Adapting the percentage intensity method to assess accelerations and decelerations in football: moving beyond absolute and arbitrary thresholds

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ABSTRACT

We adapted the percentage intensity approach to monitor accelerations and decelerations allowing players' individualisation. Forty-two players were monitored during four microcycles via global navigation satellite system devices. Raw velocity and time data were collected to calculate acceleration and deceleration magnitudes according to specific starting speed intervals, and the efforts intensities were established as very low (<25% of the maximal effort), low (25–50%), moderate (50–75%) and high (>75%). Linear regressions and Pearson correlation (r) analysed the relationship between maximal efforts and starting speeds; additionally, mean paired differences compared efforts magnitudes between subsequent starting speed intervals. Most very low intensity accelerations (86%) and decelerations (79%) started from <5 km.h^{-1}. Correlation between maximal efforts and starting speeds were $r = -0.97$ ($p < .001$) for acceleration, and $r = -0.94$ ($p < .01$) for deceleration. Maximal acceleration decreased as starting speed increases (very large effect sizes), but deceleration is less starting speed dependent (unclear to large effect sizes). This adaptation allows practitioners to individualise accelerations and decelerations classification during real-life scenarios, leading to a more precise training prescription. The very low intensity interval could be excluded to consider only relevant efforts. Maximal acceleration should be collected for each starting speed interval because accelerations are starting speed dependents.

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Introduction

During football activities, practitioners monitor players' load to assess if players are meeting the physical demands requirements and responding to the stimulus provided

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(Halson, 2014; B. R. Scott et al., 2013). Different tools can be used to this purpose, including heart rate variables and rate of perceived of effort (i.e., internal load), or variables obtained via global navigation satellite system (GNSS) (B. R. Scott et al., 2013), such as the distance covered, the displacements above a specific speed, or acceleration and deceleration occurrences (i.e., external load).

Within the different tools, the quantification of accelerations and decelerations has become widely popular, and among the most common metrics in elite football (Akenhead & Nassis, 2016). This comes as no surprise since high-intensity accelerations and decelerations (or negative accelerations) are typically associated with fatigue and exercise-induced muscle damage (Hader et al., 2019; Russel et al., 2014), with implications for post-match recovery (de Hoyo et al., 2016).

When collecting this data, counting efforts is the most-commonly used approach, followed by the distance covered accelerating or decelerating (Delves et al., 2021). To count the number of efforts, researchers and practitioners usually use scientific literature-based values, which can be very practical to use but raise two important concerns: the justification of the chosen categories and the 'one size fits all' approach. First, different 'round-figure' thresholds, such as efforts >2 $m.s^{-2}$ or >3 $m.s^{-2}$, are used in the literature but, since there is no justification to choose a specific threshold, researchers usually adopt values used in the past (McBurnie et al., 2022; Sweeting et al., 2017). Additionally, these thresholds are often associated with intensity classifications that vary across studies, making comparisons difficult (Silva et al., 2022). For example, high-intensity accelerations can differ between studies from 2–3 $m.s^{-2}$, to >2.5 $m.s^{-2}$, >3 $m.s^{-2}$ and >4 $m.s^{-2}$ (Silva et al., 2022). Secondly, those strategies disregard individual capacities. That is, if the 'high-intensity' classification is defined arbitrarily, a high-intensity action for one player could be of low- or moderate-intensity for another player (Martínez-Cabrera et al., 2021).

To address this issue, Sonderegger et al. (2016) proposed a new assessment approach, by considering the individual capacity and the impact of the effort starting speed. The authors presented the acceleration intensity calculation as the percentage of the maximal observed acceleration and the maximal voluntary acceleration that could be achieved for each starting speed [standing start; trotting (6.0 $km.h^{-1}$), jogging (10.8 $km.h^{-1}$), and running (15.0 $km.h^{-1}$)]. Then, intensity was classified as high (acceleration >75% of the maximal), moderate (>50%), low (>25%), and very low (<25%). This approach has been used by different research groups to assess accelerations (Abbott et al., 2018; De Hoyo et al., 2018; Javier Núñez et al., 2019; Martínez-Cabrera et al., 2021), with conflicting results. However, the cited papers used different starting speed thresholds and did not use the four relative acceleration intensity intervals previously proposed (Abbott et al., 2018; De Hoyo et al., 2018; Javier Núñez et al., 2019; Martínez-Cabrera et al., 2021). Nevertheless, one of the referred studies (Abbott et al., 2018), compared the absolute intensity method (where efforts are assessed with arbitrary and absolute values) with the relative intensity method during training sessions and friendly matches, reporting higher distances covered in the absolute intensity method. This means that the acceleration load can be influenced by the chosen method, altering the information collected by practitioners. Additionally, the relative approach has also been addressed to sprint displacements, with two important considerations: first, by using an individual approach, differences between playing positions were smaller that with the absolute approach (Silva et al.,

2024); and secondly, most of the registered hamstring injuries occurred >80 % of the players' maximal speed (Aiello et al., 2023). Unfortunately, since most studies that assess accelerations and decelerations demands use absolute and arbitrary thresholds (Silva et al., 2022), findings regarding relative and individual thresholds are still scarce.

Of note, in their proposal, Sonderegger et al. (2016), assessed the maximal acceleration during four acceleration tests, and assessing maximal acceleration as originally proposed could not be practical during the competitive season (De Hoyo et al., 2018). Also of note, decelerations were disregarded in the original proposal (Sonderegger et al., 2016), which is an important limitation due to the importance of these actions in football practices (Harper & Kiely, 2018). However, this promising proposal presents two major advantages in comparison to the absolute approach: first, by using individual and relative thresholds, players capabilities are considered, which do not occur with absolute and arbitrary approaches; and second, the authors acknowledged that the accelerations starting speed influences the acceleration capacity. Regarding this second advantage, it is fundamental to highlight the importance of the starting speed in acceleration efforts. For example, considering an acceleration of >2.5 $m.s^{-2}$ starting at 9 $km.h^{-1}$ can more than double the metabolic cost when compared to a constant displacement at the same speed (9 $km.h^{-1}$) (Osgnach et al., 2010). Also, this individual acceleration-speed profile—where the acceleration magnitude decreases with increases in the starting speed—was also reported in a more recent study (Morin et al., 2021).

Therefore, considering the merits of the relative approach, we aim to provide an adaptation to this method, by assessing maximal efforts during football activities (instead of tests) and therefore facilitating the inclusion of decelerations' assessment with this method. We hypothesised that acceleration magnitudes would decrease with starting speed increases, and deceleration magnitudes would increase with starting speed increases.

Materials and methods

Procedures

Forty-two male professional football players from two Portuguese teams competing in the first division were monitored during training sessions from one full mesocycle (4 weeks). Data were collected in the 2020/21 and 21/22 seasons (one team per season).

Sample

Coaching staff monitored and collected the daily training data from forty-two male professional players. To be included in this study, a minimum of 50% participation in training sessions was required. The average age, height and weight were 26.7 ± 4.2 years, 181.7 ± 6.3 cm, and 74.5 ± 6.0 kg. Due to the particularities of the position, goalkeepers were excluded from data collection. Sample characteristics are presented in Table 1. Ethics committee approval (35/2021) and written consent from the clubs were obtained. Data analysis was performed anonymously.

Table 1. Sample characteristics as number of players, training files and average training sessions attendances (mean ± SD) per playing position.

	CD	FB	CM	WM	FW	Total
Players (*n*)	8	10	13	6	5	42
Training files (*n*)	148	192	264	110	98	812
Percentage of training sessions attendance (%)	84.48 ± 12.25	89.08 ± 8.85	92.31 ± 7.37	84.58 ± 6.59	90.33 ± 10.26	88.71 ± 9.64

CD = central defender; FB = fullback; CM = central midfielder; WM = wide midfielder; FW = forward.

Acceleration and deceleration assessment

The teams monitored their players using a 10-Hz global positioning system (Catapult Vector S7 and Vector X7 - one model per team—Catapult Sports, Melbourne, Australia) that encompassed a double constellation system (GNSS and GPS). Both models are FIFA certified (*FIFA Resource Hub - Catapult*, n.d.) and the 10-Hz sampling rate has been validated to assess accelerations in team-sports (M. Scott et al., 2016). Devices were secured between the upper scapulae, at approximately the T3–4 junction and were activated 15 minutes before use, in accordance with the manufacturer's instructions.

Raw data of training sessions were retrieved from the proprietary software (OpenField Console, Catapult Sports, Melbourne, Australia) and the velocity ($m.s^{-1}$) and time (m.s) were used to calculate accelerations and decelerations ($m.s^{-2}$). Maximal accelerations and decelerations were retrieved as the maximal individual value when another effort was registered within 1 $m.s^{-2}$ (for example, a maximal acceleration of 5.2 $m.s^{-2}$ required another acceleration ≥4.2 $m.s^{-2}$; the same procedure for decelerations). Training procedures were conducted as usual with no interference from the research team.

Thresholds and intensities

Players' acceleration and deceleration efforts from training sessions were retrieved with the starting speed ($km.h^{-1}$) and the effort magnitude ($m.s^{-2}$). The starting speed was retrieved as the registered speed when the speed started increasing (acceleration) or decreasing (deceleration). Accelerations and decelerations were calculated as the change of speed divided by the change of time—as long as speed kept increasing or decreasing, one effort was counted. As so, no minimum effort duration was applied to ensure that all efforts were accounted (Delves et al., 2021).

For the percentage intensity calculation, we adapted the Sonderegger et al. (2016) proposal, using the following equation to calculate percentage acceleration:

$$\text{Percentage acceleration} = \frac{\text{amax, action}}{\text{amax}} * 100$$

Where 'amax, action' corresponds to the action acceleration or deceleration (each individual effort) according to the starting speed bandwidth interval (<5 $km.h^{-1}$, 5–10 $km.h^{-1}$, 10–15 $km.h^{-1}$, 15–20 $km.h^{-1}$, 20–25 $km.h^{-1}$, 25–30 $km.h^{-1}$ and >30 $km.h^{-1}$); 'amax' corresponds to the maximal acceleration or deceleration for each starting speed

bandwidth interval (<5 km.h^{-1}, 5–10 km.h^{-1}, 10–15 km.h^{-1}, 15–20 km.h^{-}1, 20–25 km.h^{-1}, 25–30 km.h^{-1} and >30 km.h^{-1}).

Accelerations and decelerations intensities were categorised as high (>75%), moderate (25–50%), low (25–50%), and very low (<25%) as proposed by Sonderegger et al. (2016).

Statistical analysis

Descriptive statistics were conducted to analyse efforts intensities and starting speeds. Accelerations and decelerations occurrences were analysed as a percentage of all occurrences. Linear regressions and Pearson correlation (r) were calculated to analyse the relationship between the maximal acceleration and decelerations and starting speeds. Mean paired differences compared accelerations and decelerations magnitudes between subsequent starting speed intervals. Cohen's (d) effect sizes were established as trivial (<0.2), small (0.2 < 0.6), moderate (0.6 < 1.2), large (1.2 < 2.0), very large (2.0 < 4.0) and huge (>4.0) with 90% confidence intervals (Hopkins et al., 2009). If the CI crossed zero, an unclear effect size was established (Batterham & Hopkins, 2006). Additionally, a Linear Mixed Model, as they extend standard linear regression models through the introduction of random effects and/or correlated residual errors, was conducted with maximal accelerations/decelerations as dependent variable, initial velocity of the effort as fixed factor and player individuality and player position as random factor. Given that the Linear Mixed Model, which is one of the Gaussian models, is generally robust to violations of the normality assumption (Knief & Forstmeier, 2021), it was not necessary to test the normality of the data. All the statistical analysis were performed in jamovi with the ESCI package (Jamovi, n.d.; Team, 2021)

Results

Most of very low intensity accelerations and decelerations started from <5 km.h^{-1} (86% and 79% respectively). Similarly, most of low and moderate intensity accelerations started from <5 km.h^{-1} (79% and 68% respectively), while most of low and moderate intensity decelerations started between 5–15 km.h^{-1} (66% and 61%, respectively). High-intensity accelerations started mainly <10 km.h^{-1} (70%), while high-intensity decelerations varied from starting speeds of 15–20 km.h^{-1} (19%), 5–10 km.h^{-1} (19%), and 10–15 km.h^{-1} (31%). This is represented in Figure 1.

Means ± SD of maximal and average accelerations and decelerations per starting speed interval are presented in Table 2.

Correlation between maximal accelerations and decelerations were −0.97 ($p < .001$) and −0.94 ($p < .01$), respectively. Linear regressions and respective equations are presented in Figure 2.

As starting speed increased, differences in maximal accelerations decrease with very large effect sizes. The only exception was when comparing 20–25 km.h^{-1} with 25–30 km.h^{-1} starting speed intervals (large effect size). Maximal decelerations decrease between starting speed intervals, except between <5 km.h^{-1} and 5–10 km.h^{-1} (unclear), and between 10–15 km.h^{-1} and 15–20 km.h^{-1} (maximal decelerations increased with a small effect size). These differences are represented in Figure 3.

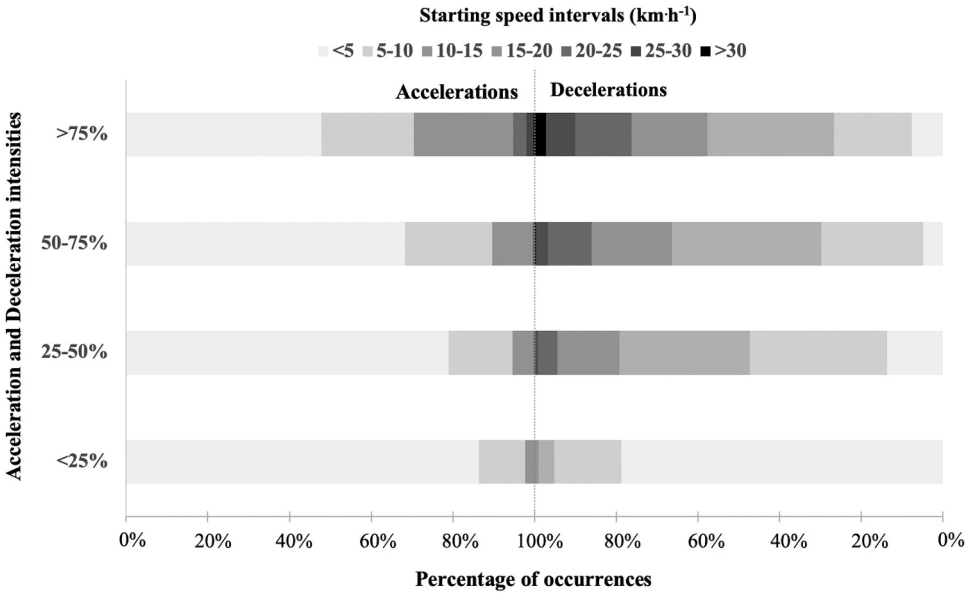

Figure 1. Comparison of mean percentage of accelerations and decelerations occurrences per training session separated per intensity interval, and according to starting speed. Acceleration and decelerations intensities (<25%, 25–50%, 50–75%, and > 75%) refer to the relative intensity according to each player peak effort according to each starting speed interval (the speed at which the effort started).

Table 2. Mean ± SD of maximal and average magnitudes of accelerations and decelerations for each bandwidth starting speed interval.

	Acceleration		Deceleration	
Starting Speed Interval	Maximal ($m.s^{-2}$)	Average ($m.s^{-2}$)	Maximal ($m.s^{-2}$)	Average ($m.s^{-2}$)
<5 $km.h^{-1}$	4.52 ± 0.66	0.36 ± 0.46	6.08 ± 2.93	0.23 ± 0.26
5–10 $km.h^{-1}$	3.11 ± 0.31	0.29 ± 0.41	5.44 ± 3.34	0.43 ± 0.54
10–15 $km.h^{-1}$	2.10 ± 0.28	0.29 ± 0.35	4.64 ± 0.66	0.89 ± 0.78
15–20 $km.h^{-1}$	1.43 ± 0.25	0.31 ± 0.31	5.00 ± 0.96	1.36 ± 0.88
20–25 $km.h^{-1}$	0.80 ± 0.24	0.24 ± 0.22	4.41 ± 0.79	1.61 ± 0.92
25–30 $km.h^{-1}$	0.45 ± 0.26	0.23 ± 0.22	3.56 ± 0.72	1.72 ± 0.88
>30 $km.h^{-1}$	0.14 ± 0.07	0.14 ± 0.06	2.24 ± 0.66	1.63 ± 0.65

Starting speed interval refers to the speed that an acceleration or deceleration started.

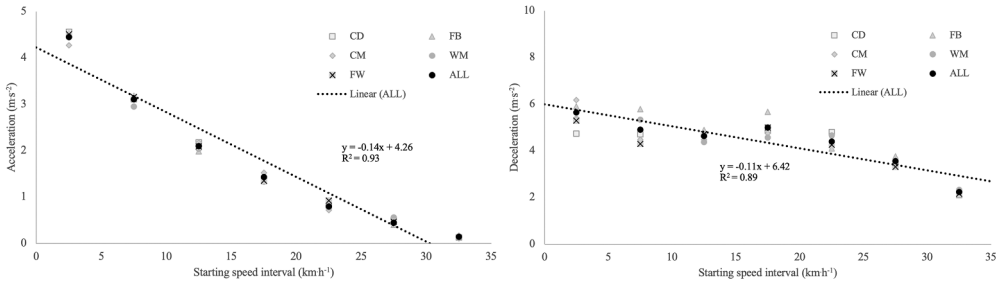

Figure 2. Linear regression of maximal accelerations (ACC) and decelerations (DEC) for starting speed bandwidth (<5 $km.h^{-1}$, 5-10 $km.h^{-1}$, 10-15 $km.h^{-1}$, 15-20 $km.h^{-1}$, 20–25 $km.h^{-1}$, 25–30 $km.h^{-1}$ and >30 $km.h^{-1}$) for all players ($n = 42$) and for each playing position. CD = central defenders; FB = fullbacks; CM = central midfielders; WM = wide midfielders; FW = forwards. Accelerations and decelerations are presented with magnitudes. The starting speed interval refers to the speed at which the effort (acceleration or deceleration) started.

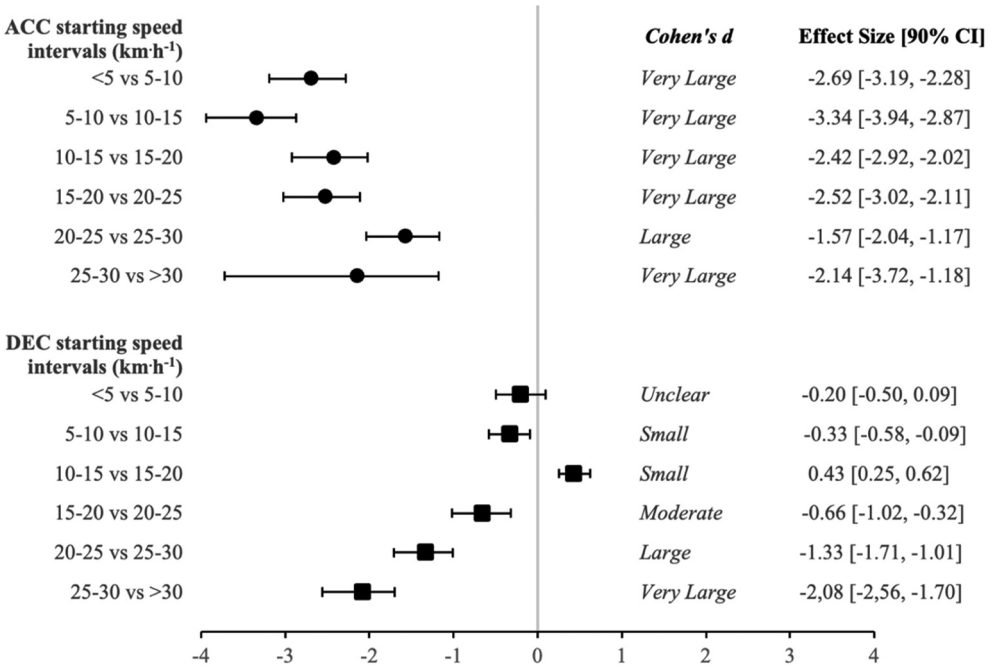

Figure 3. Effect sizes (Cohen's d) of the differences between starting speed intervals for maximal accelerations (ACC) and decelerations (DEC). Each starting speed interval (where the acceleration or the deceleration started) is compared with subsequent interval.

The linear mixed model showed that accelerations decrease with the increase of the initial velocity ($F = 1085.81$, $p < .001$, $R^2 = 0.96$), regardless of the players' positions (ICC = 0.00; Variance = 0.00) and of the players' individuality (ICC = 0.08; Variance = 0.01). However, decelerations are less dependent of the initial velocity ($F = 35.08$, $p < .001$, $R^2 = 0.42$), regardless of the players' positions (ICC = 0.00; Variance = 0.00) and of the players' individuality (ICC = 0.03; Variance = 0.04). This means that the initial velocity (fixed factor) explained almost all variance registered in accelerations but not in decelerations.

Discussion and implications

The main purpose of this study was to provide a new method to classify acceleration and deceleration intensities during football training, by adapting the individual percentage intensity method proposed by Sonderegger et al. (2016). We presented a practical method that allows practitioners and researchers to classify players' individual accelerations and decelerations efforts during training sessions and that can be extended to competitive matches. In fact, since we used the maximal acceleration and deceleration as the highest values obtained during training sessions (per starting speed bandwidth interval), one can choose to use maximal values from each training session, from the full microcycle (highest value of the week) or from the week match, without needing field tests. This is important because field tests can only provide a glimpse of players' status, as players change their capacities across the season and constant testing is difficult to

implement in elite contexts (Lovell et al., 2019). Additionally, these tests are usually performed with a standing start which differs from what happens in competition and training practice. By assessing maximal values during practice, decelerations can also be classified.

With our adaptation proposal, three main outcomes should be highlighted: very low intensities can be excluded from future analysis; efforts starting >30 km.h^{-1} were practically inexistent; the efforts' starting speed should be consider only to establish acceleration, but not deceleration, thresholds. Regarding the very low intensity threshold (efforts <25% of the maximal effort), we found that most of these efforts started from speeds <5 km.h^{-1} (Figure 1) which means that these efforts probably represent insignificant changes of velocity and would probably relate to noise. Since we used raw data, we should acknowledge the risk of including noise in the data. For example, accelerations <1 m.s^{-2} are usually excluded from analysis (Silva et al., 2022), which did not occur in this study. However, we purposely did not exclude that data from the analysis, because acceleration capacity decreases as starting speed decreases (Figure 2) (Sonderegger et al., 2016). As so, excluding accelerations <1 m.s^{-2} could exclude efforts that, due to elevated starting speeds, were moderate or even intense. Although excluding accelerations with specific magnitudes can exclude important data, by considering the starting speed, we can recommend practitioners to disregard this intensity interval (<25%) because they probably represent a negligible mechanical load to the player and can carry movement 'artifacts'. As for the efforts starting speed bandwidth intervals, we found that very few efforts occur in the highest interval (>30 km.h^{-1}), which can condition maximal values. For example, if only two accelerations occur in this interval, one would be the maximal and the other would have a high impact when collecting load, by representing 50% of the load reported in that starting speed interval. Furthermore, 25 km.h^{-1} is a threshold very similar to the most used sprint threshold (Gualtieri et al., 2023), which is the highest speed threshold used. As so, we also recommend merging the starting speed intervals of 25–30 km.h^{-1} and >30 km.h^{-1} to >25 km.h^{-1}. Finally, and considering the influence of the starting speed on accelerative and decelerative capacities, we found that decelerations are somehow independent from the starting speed. This means that to classify decelerations intensity, practitioners do not need to consider starting speed. This may seem counterintuitive, when reflecting on deceleration testing procedures: athletes accelerate for 20 metres, and then perform a maximal deceleration and back-pedal to the 20 metres line (where deceleration started) (Harper et al., 2020). However, during training sessions, players can instinctively avoid very high intense decelerations since these are associated with elevated injury risk (Harper & Kiely, 2018). This does not mean that we should expect maximal decelerations (highest magnitude) from low starting speeds (<5 km.h^{-1}). Instead, it means that the highest deceleration magnitude can be registered in different starting speeds intervals. Furthermore, the technology used in this test is still scarce in football, but monitoring deceleration loads is crucial as incorrect volume and intensity exposure can lead to tissue damage and increased injury risk (Harper & Kiely, 2018).

This partially aligns with our hypothesis. That is, players achieve lower acceleration magnitudes at higher starting speeds, but during the different starting speeds intervals, players achieve higher decelerations magnitudes. This is probably because players perform sudden stops from lower speeds, while slowly decelerate when at higher starting speeds, protecting themselves. As so, it seems that the acceleration capacity is more starting speed dependent

than the deceleration capacity—as seen with the average accelerations and decelerations for each starting speed bandwidth interval, and when comparing the maximal efforts between subsequent starting speed intervals (Figure 3).

We presented an adaptation from an interesting proposal to move behind absolute thresholds, which have several limitations as previously described (McBurnie et al., 2022; Silva et al., 2022; Sweeting et al., 2017). Interestingly, two studies compared this arbitrary classification with an individualise classification. First, Abbott et al. (2018) compared a global method (low intensity: 1–2 $m.s^{-2}$; moderate intensity: 2–3 $m.s^{-2}$; and high intensity: >3 $m.s^{-2}$) with an individualise method during training sessions and friendly matches (low intensity: 25–50%; moderate intensity: 50–75%; and high intensity: >75%) and calculated the maximum acceleration with players' testing. Players were then divided into groups according to their acceleration capacity (low, medium, and high). The authors reported higher distances covered in the global method (medium group: moderate and high-intensity accelerations; high group: all intensities). Hence, how practitioners classify players accelerations and decelerations impacts the load reported. Secondly, Fischer-Sonderegger et al. (2019) compared their percentage intensity method (Sonderegger et al., 2016) with a speed-based method (measured the distance cover in different speed zones: standing, 0.0–0.7 $km.h^{-1}$; walking, 0.7–7.2 $km.h^{-1}$; jogging, 7.2–14.4 $km.h^{-1}$; running, 14.4–19.8 $km.h^{-1}$; high-speed running, 19.8–25.2 $km.h^{-1}$; and sprinting, >25.2 $km.h^{-1}$) and with an absolute method (low intensity: 1–2 $m.s^{-2}$; moderate intensity: 2–3 $m.s^{-2}$; high intensity: >3 $m.s^{-2}$; and another high-intensity interval: >4 $m.s^{-2}$). However, the absolute threshold method was also dependent from starting speed and the statistical analyses focused on the sample level. From the latter study, differences were reported between methods, but similarities can be found in this study such as the number of efforts in >75% and >4 $m.s^{-2}$ (29.7 vs. 30.7) intervals with starting speeds of 0–1 $m.s^{-1}$. Finally, the latter study used open-ended thresholds (>3 $m.s^{-2}$ and >4 $m.s^{-2}$), which lead to the inclusion of different thresholds in the same interval (Varley et al., 2017).

One additional consideration refers to playing positions. Although, differences between playing positions are expected with absolute thresholds (Silva et al., 2022), the relative method accounts for that individuality. As seen in Figure 2, the maximal effort for each starting speed was similar across the different positions. Reduced differences between playing positions within relative approaches were also reported when comparing sprint displacements in elite soccer players (Silva et al., 2024). This is somehow expected, because if a player reaches lower speeds or acceleration/deceleration magnitudes during real-scenario (such as training sessions or competitive matches) due to their position characteristics, then that would be reflected in an individualised approach such as the one presented in this study. Since we have focussed on acceleration and deceleration intensities, no players' categorisation (such as fast or slow players) was applied. However, this information could be interesting in future research.

We assessed maximal accelerations and decelerations efforts during training sessions, instead of performing field tests. Although it can be questioned if the maximal efforts represent players' maximal capacities, we considered four microcycles to collect maximal efforts. As so, using only one match or one training session could mislead practitioners regarding maximal values. However, field tests can also mislead practitioners with players failing to replicate their field tests maximal velocities during matches (Al Haddad et al., 2015; Djaoui et al., 2017). Other limitation relates to our sample consisting only of training sessions, while it would also be important to consider competition. Future studies are necessary to

extend our findings to prolonged periods of observation, and inclusion of competitive matches. However, we stress that this study aimed to adapt the original method proposed by Sonderegger et al. (2016) so it can be considered a methodological study.

As so acceleration and deceleration monitoring should be individualised and not arbitrary, with intensity efforts representing a percentage of the maximal individual effort. With this, practitioners avoid the 'one size fits all' problem, where a 'high-intensity' effort for one player, by being defined arbitrarily, can be a low- or moderate-intensity for another (Martínez-Cabrera et al., 2021). Additionally, when assessing accelerations intensity, practitioners should assess the maximal acceleration for each starting speed bandwidth interval—as acceleration capacity diminishes as the starting speed increases; but when assessing decelerations intensity, practitioners should assess the maximal deceleration regardless of the starting speed—as deceleration capacity is less dependent on the effort starting speed. Finally, very-low intensity efforts (<25%) should be excluded from overall acceleration and deceleration demands, because these efforts potentially represent insignificant velocity changes, occurring mostly (~80%) from <5 $km.h^{-1}$ starting speeds.

Conclusion

We presented an adaptation of the percentage intensity approach, by assessing maximal accelerations and decelerations from training data. To assess maximal values, we recommend consider the maximal acceleration for each starting speed interval, and the overall maximal deceleration, independent of starting speed intervals. This is justified by the higher dependence of acceleration than deceleration to the starting speed. Finally, considering low, moderate and high intensities would provide the fundamental information to practitioners, without excluding relevant efforts. This strategy can be used to replace the absolute threshold method, with two major advantages: individualisation of players' capacities/demands and ability to assess players according to their physical performance during real-life scenario such as training sessions and/or matches.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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